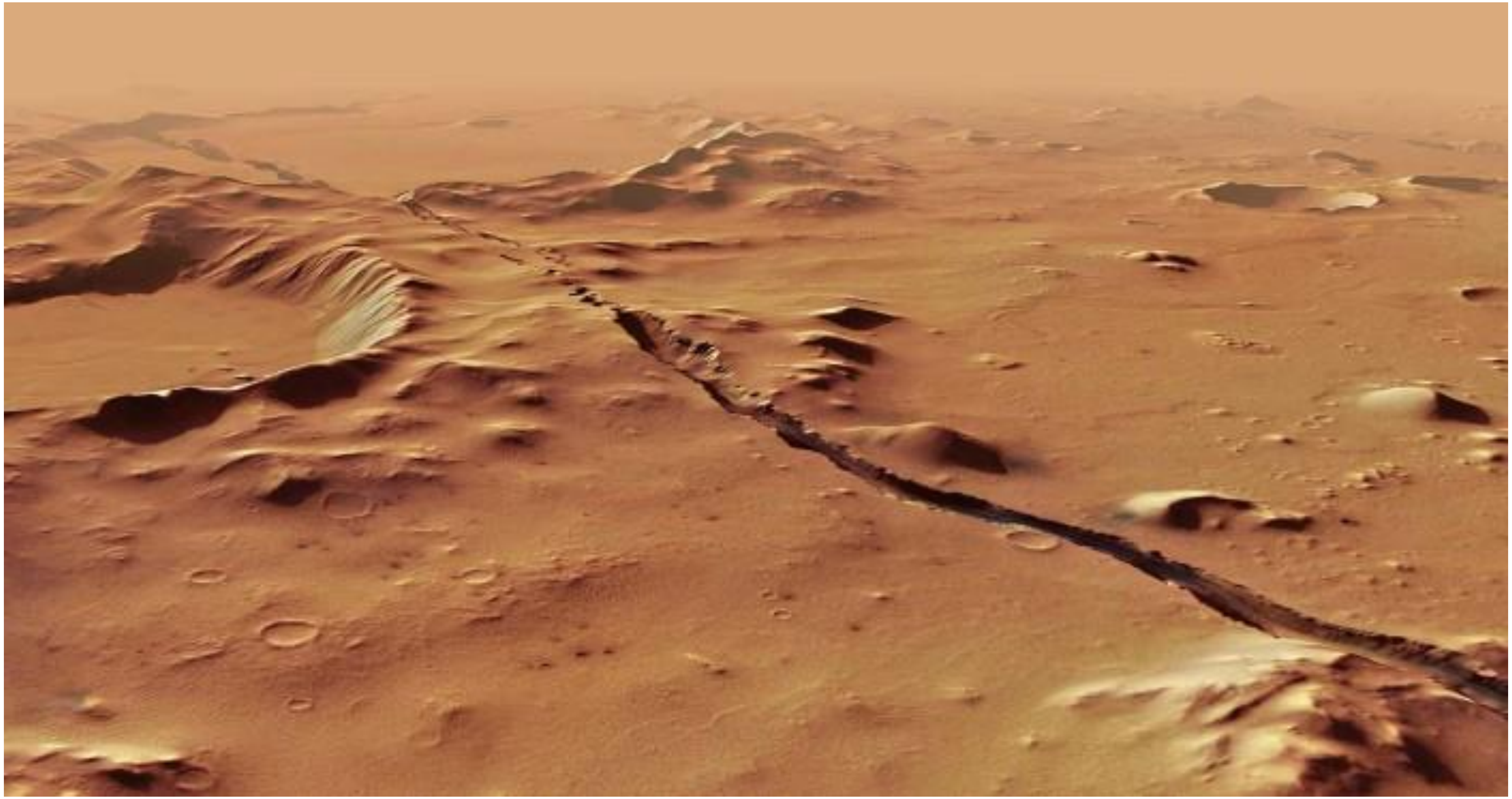
Geomorphology and Geophysical signatures observed by various missions



Prof. Giri Yellalacheruvu, AGP

Why we need to explore/study other planets?

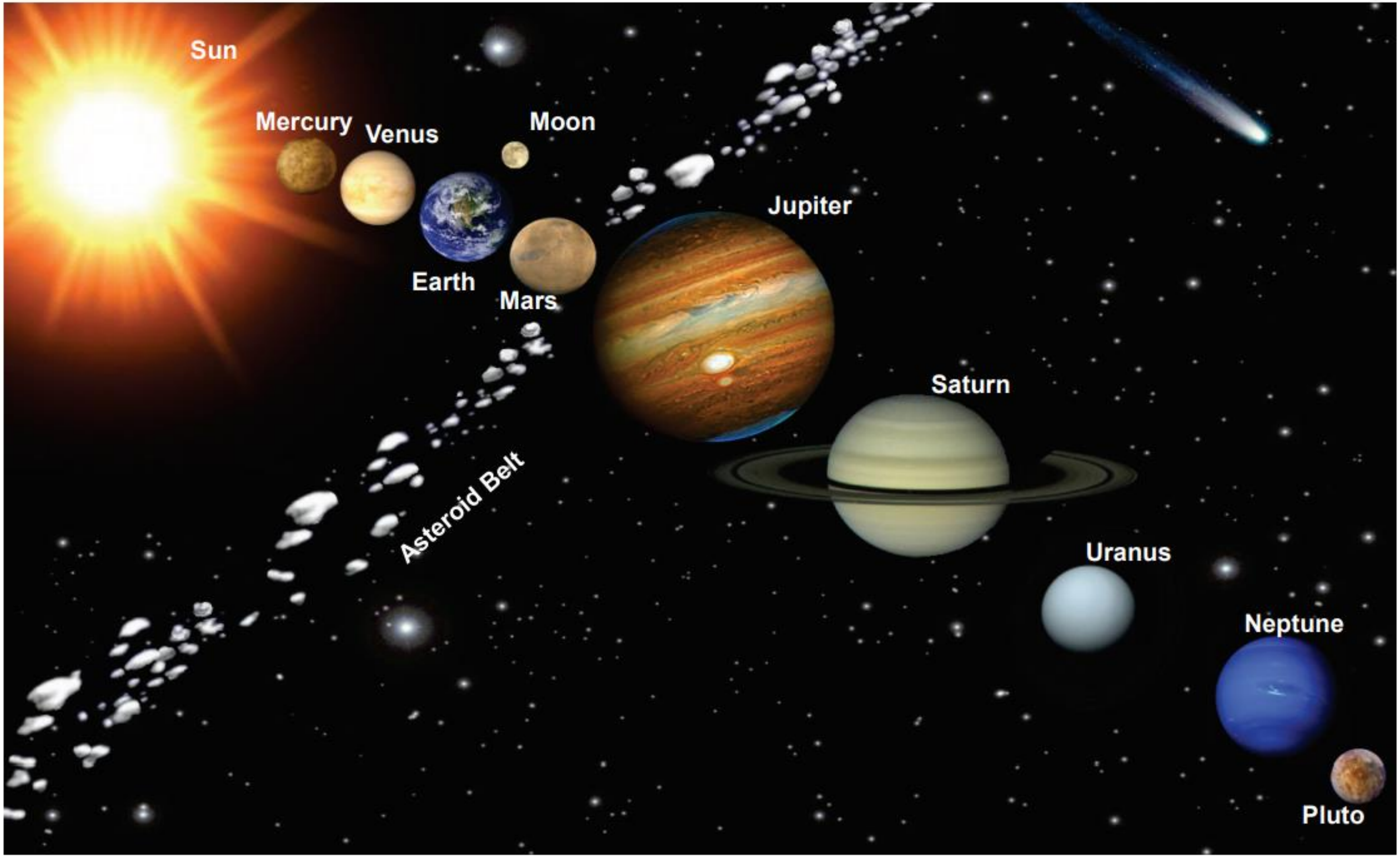
What Planets/Moons we are interested and why??

How we can know about the distant Planets/Moons??

Is these Planets and Moons are evolved in a similar fashion like our mother Earth!

What the rocks present on the surface of these Planets/Moons tells us?

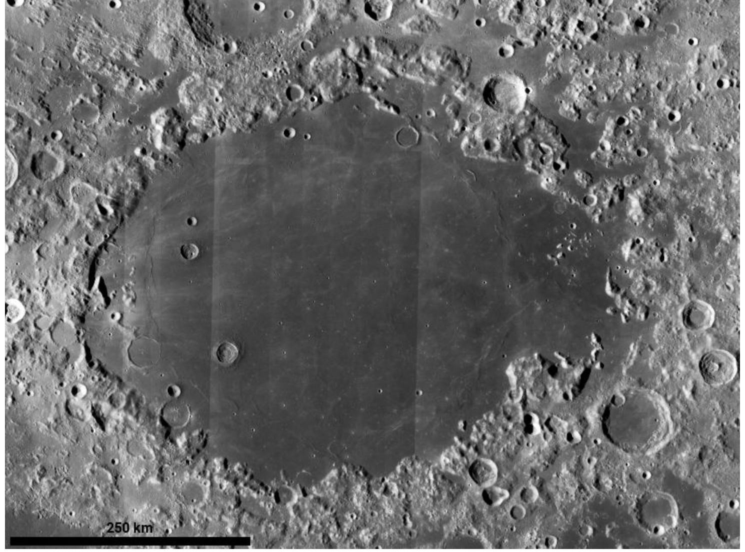
What causes Moonquakes and Marsquakes?



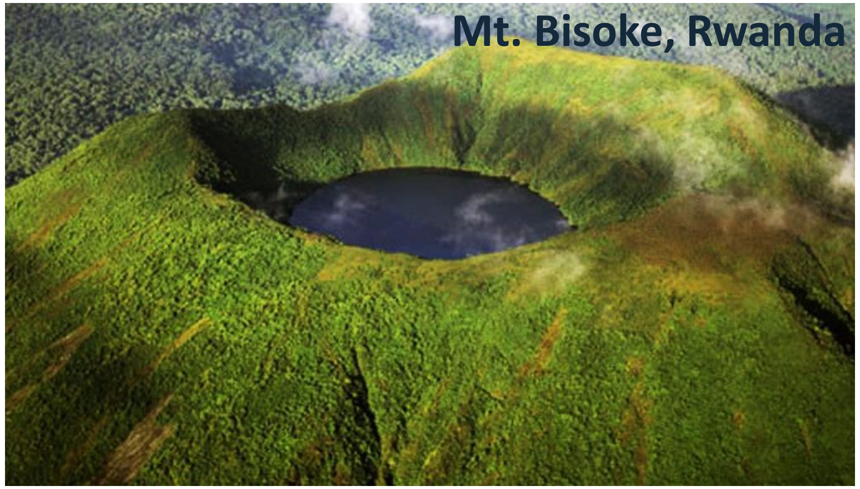
An artistic view of our solar system.

Source: https://spacetechinfo.worldpress.com

- **Mare:** any flat, dark plain of lower elevation on the Moon.

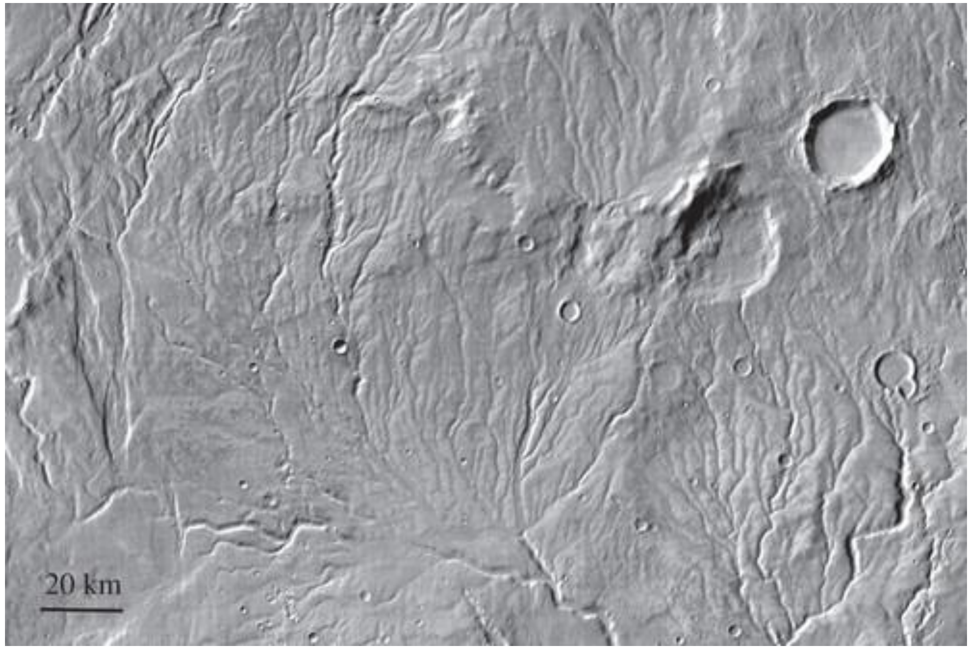


Impact crater: This is formed when an object like an asteroid or meteorite crashes into the surface of a larger solid object like a planet or a moon

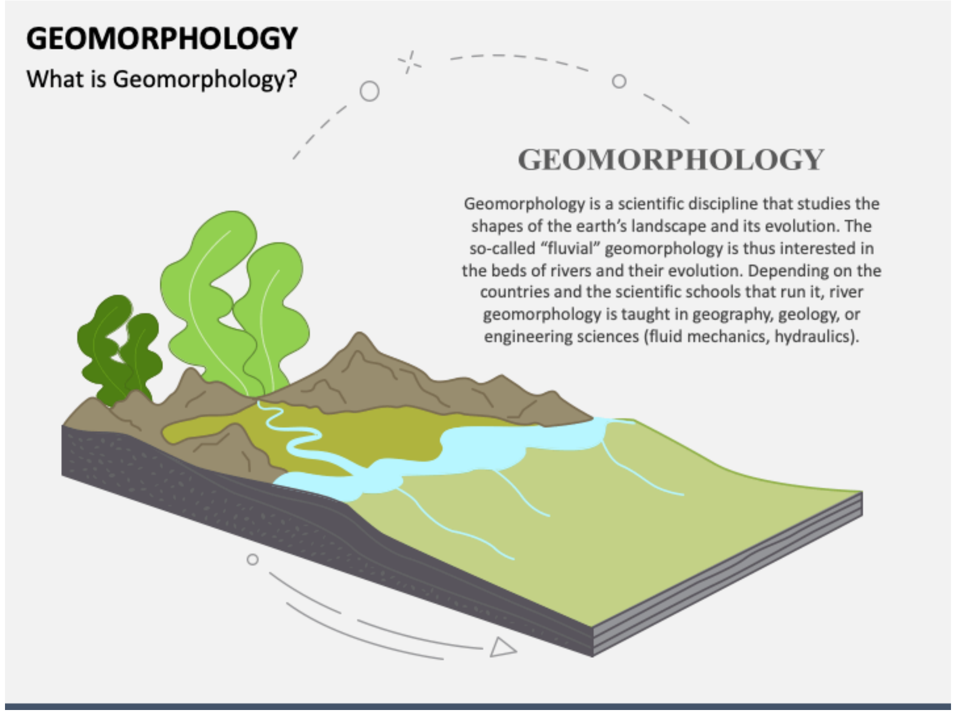


- **Volcanic crater:** This is a bowl-shaped depression produced by the impact of a meteorite, volcanic activity, or an explosion.

What Proxies we are looking for??



Warrego Vallis at 42° S, 267° E, one of the most densely dissected areas of the planet. The drainage density strongly supports precipitation and surface runoff



Kasei Vallis (25° N, 300° E), the largest outflow channel on the planet, emerges from a shallow north–south canyon to the west of the scene shown here. To the east, the channel extends deep into the northern plains

Luna 1	USSR	Jan. 2, 1959	Jan. 4, 1959	Impact	Partial Success; first Moon flyby
Pioneer 4	USA	March 3, 1959	Mar. 4., 1959	Flyby	Partial Success
Unnamed Luna	USSR	June 18, 1959	N/A	Impact	Unsuccessful
Luna 2	USSR	Sept. 12, 1959	Sept. 13, 1959	Impact	Successful; first spacecraft to impact the Moon's surface
Pioneer P-1	USA	Sept. 24, 1959	N/A	Orbiter	Unsuccessful
Luna 3	USSR	Oct. 4, 1959	Oct. 6, 1959	Flyby	Successful; first pictures of the lunar farside.
Pioneer P-3	USA	Nov. 26, 1959	N/A	Orbiter	Unsuccessful

Name	**Nation**	**Launch**	**Arrival**	**Type**	**Outcome**
SMART-1	Europe	9/27/2003	11/15/2004	Orbiter/Impact	Successful; first European Moon mission
SELENE (Kaguya)	Japan	9/14/2007	10/3/2007	Orbiter/Impact	Successful
Chang'e 1	China	10/24/2007	11/5/2007	Orbiter/impactor	Successful; first Chinese Moon mission
Chandrayaan-1	India	10/22/2008	11/12/2008	Orbiter	Successful
Lunar Reconnaissance Orbiter (LRO)	USA	6/18/2009	6/23/2009	Orbiter	(Active Mission) Successful; extended mission in progress
LCROSS	USA	6/18/2009	10/9/2009	Impact	Successful; impact of LRO upper stages

1950s: Dawn of the Space Age

1960s: Race to the Moon

1970s: Sampling the Moon

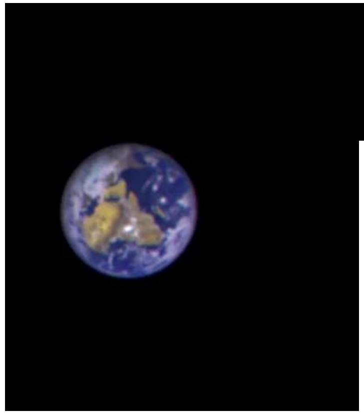
1980s: Quiet Moon

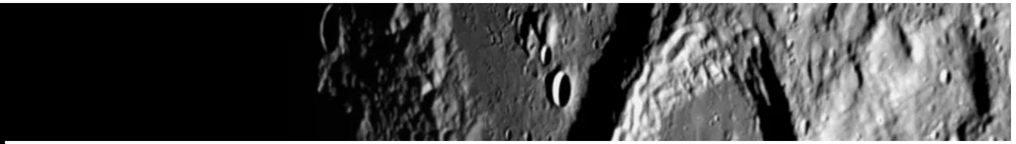
1990s: Robots Return

2000s: International Moon

2010s: Delving Deeper

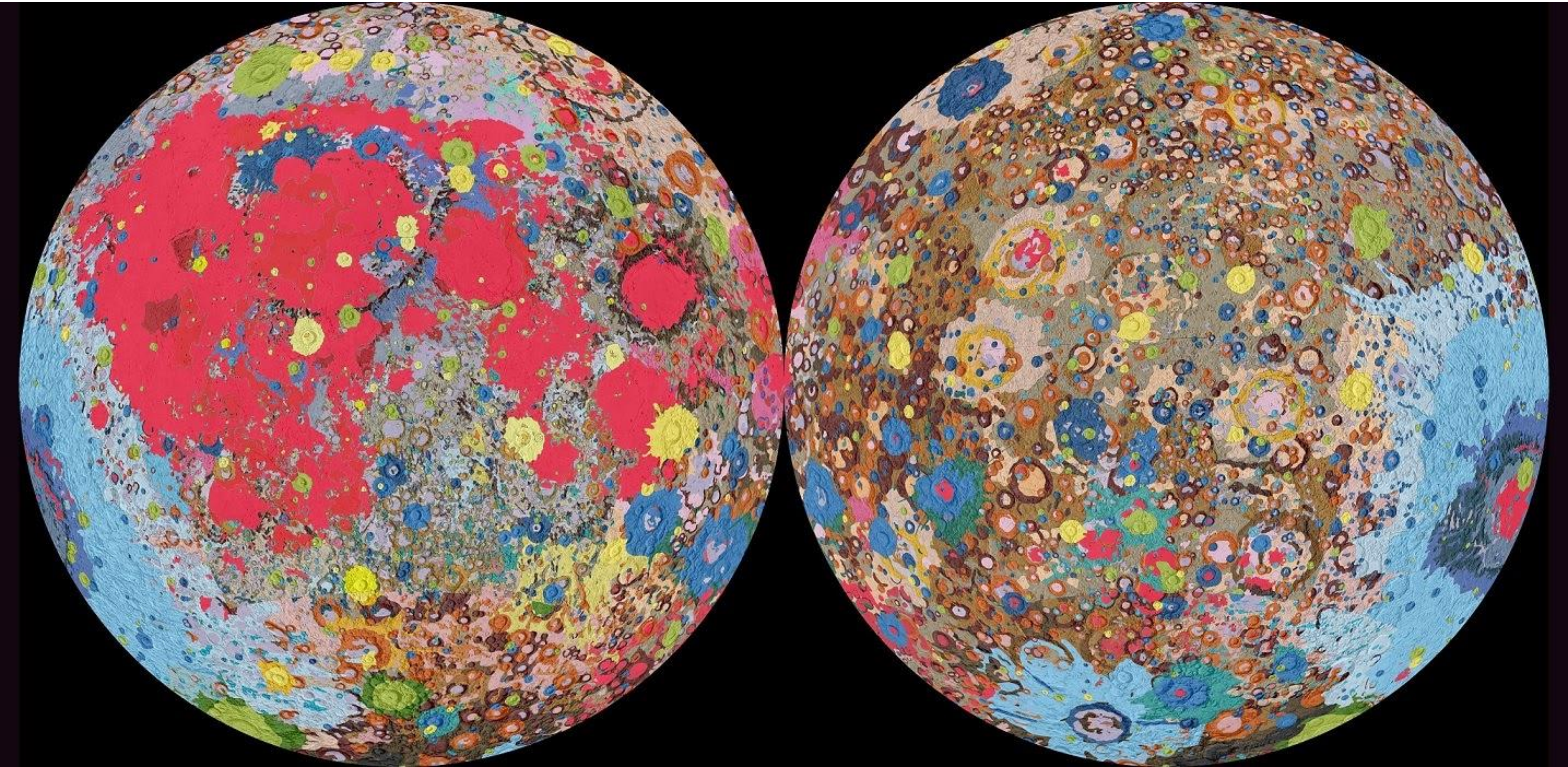
2020s: Back to the Surface



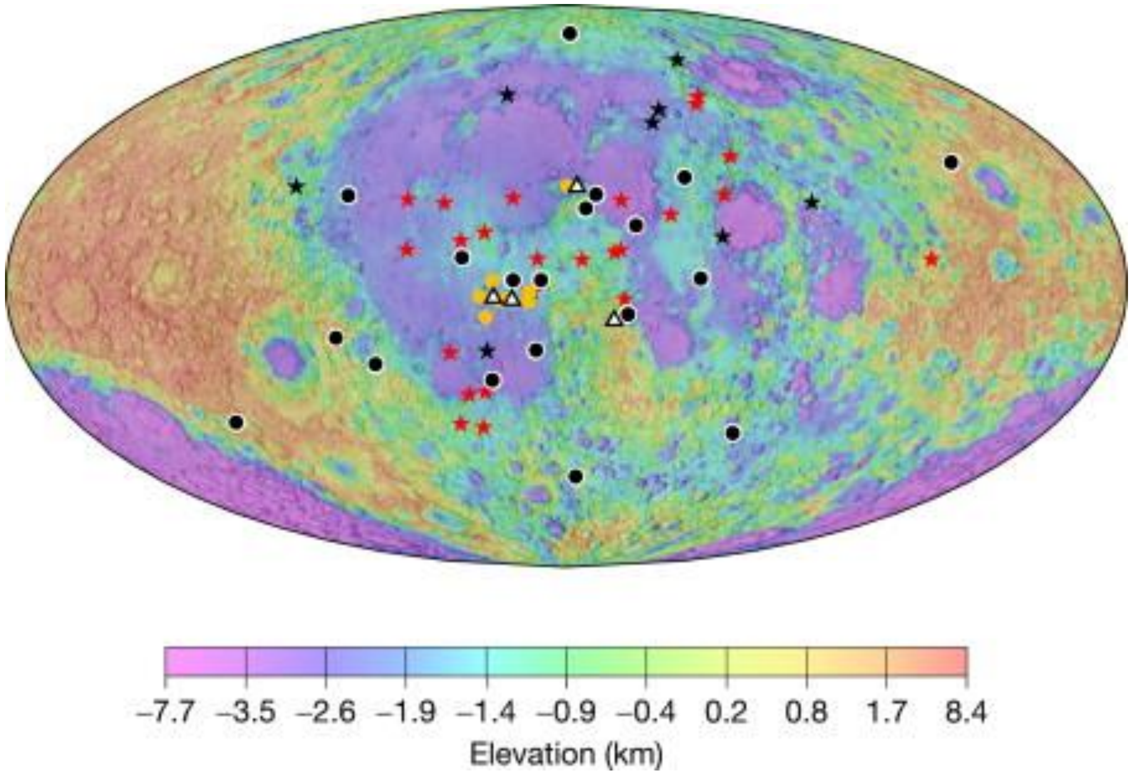


Name	Nation	Launch	Arrival	Type	Results
Chang'e 5	China	11/23/2020	12/1/2020	Sample Return	Successful
Danuri (Korean Pathfinder Lunar Orbiter)	South Korea	8/04/2022	12/16/2022	Orbiter	Successful
Artemis I	USA	11/16/2022	11/21/2022	Flyby	Successful; first flight test of the Space Launch System rocket and the Orion capsule
Hakuto-R Mission 1	Japan (ispace)	12/11/2022	3/21/2023	Lander	Unsuccessful
Emirates Lunar Mission	United Arab Emirates	12/11/2022	3/21/2023	Rover	Unsuccessful
Chandrayaan-3	India	07/14/2023	08/23/2023	Lander and Rover	Successful; First spacecraft to soft land near the lunar South Pole.

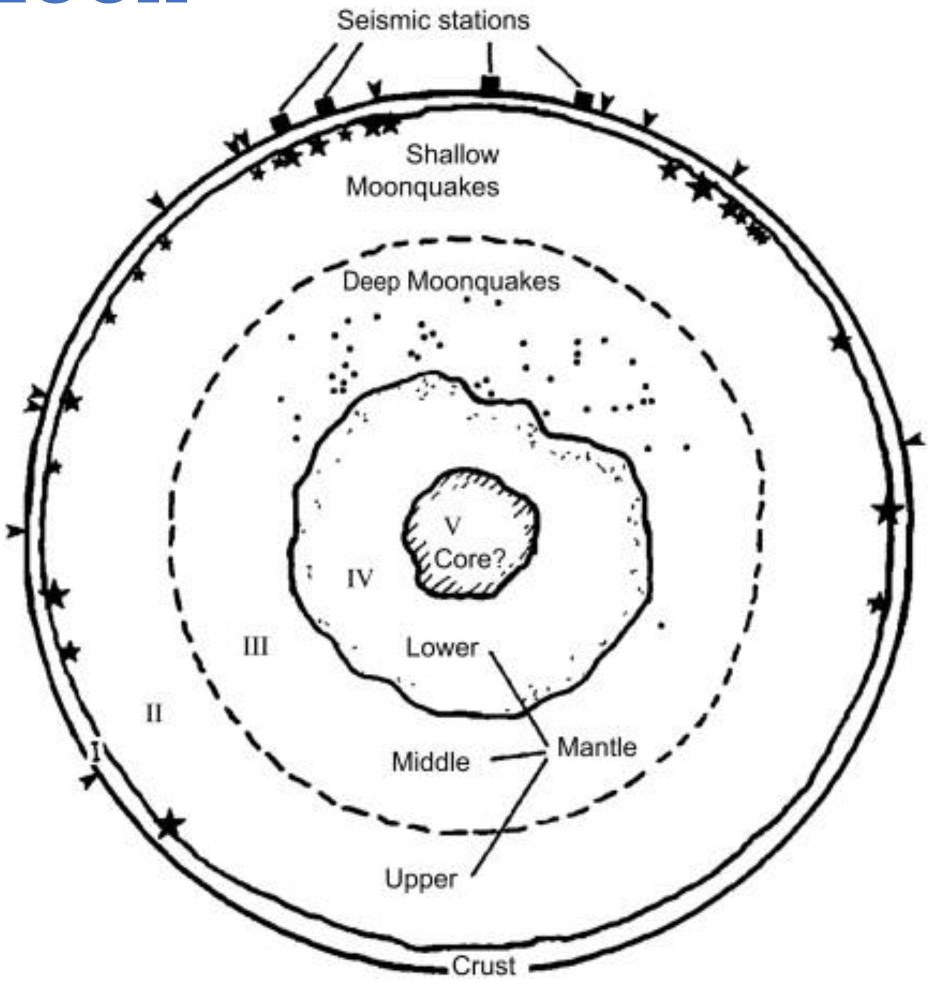
Geology of Moon



Seismicity on Moon



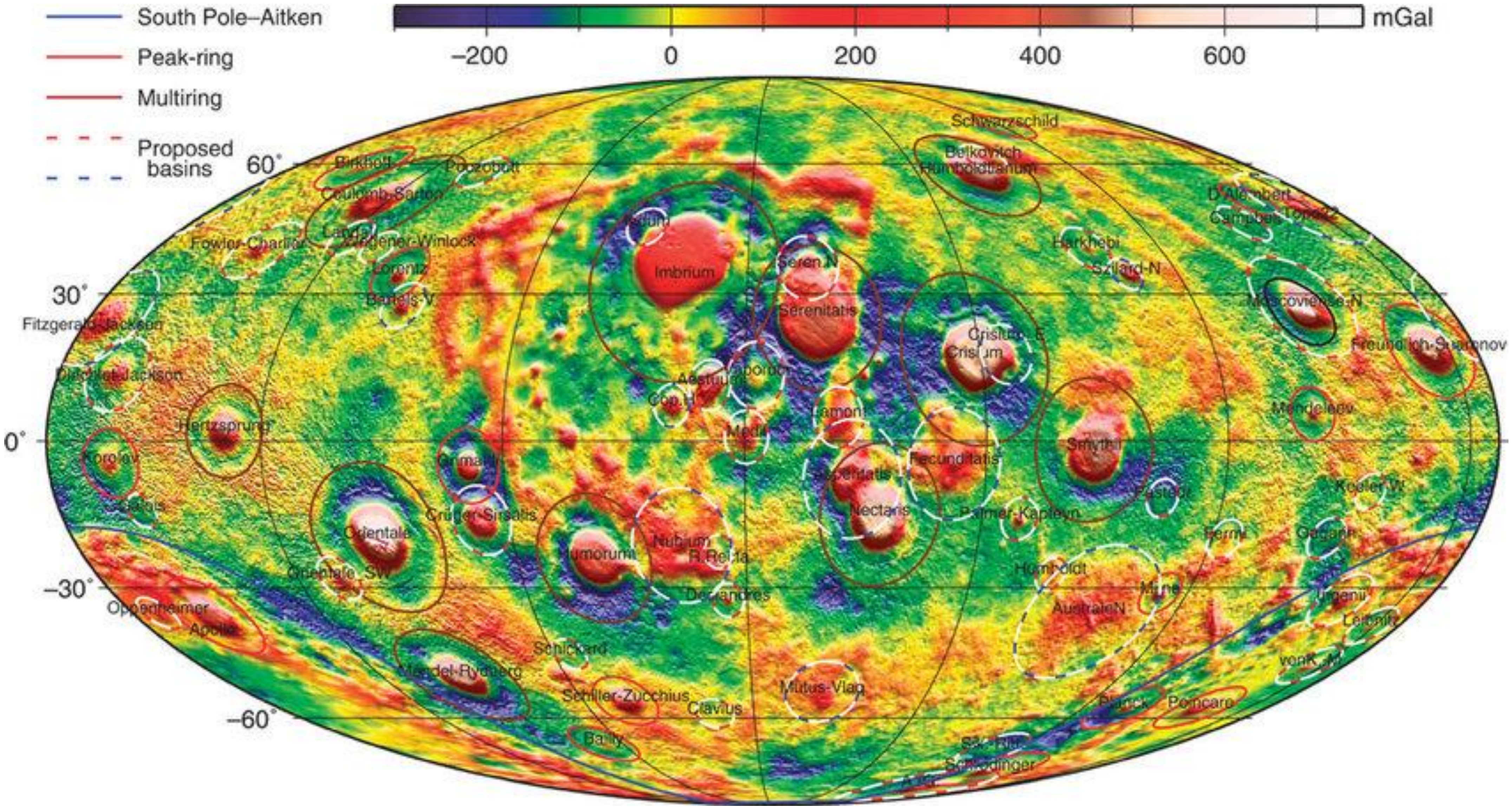
The figure shows Apollo stations 12, 14, 15, and 16 (white triangles), along with the locations of 24 deep-event clusters (red stars), 8 shallow events (black stars), 19 meteoroid impacts (black circles), and 8 artificial impacts (orange circles). The background color map is the Clementine topography map (USGS map I-2769, 2002).



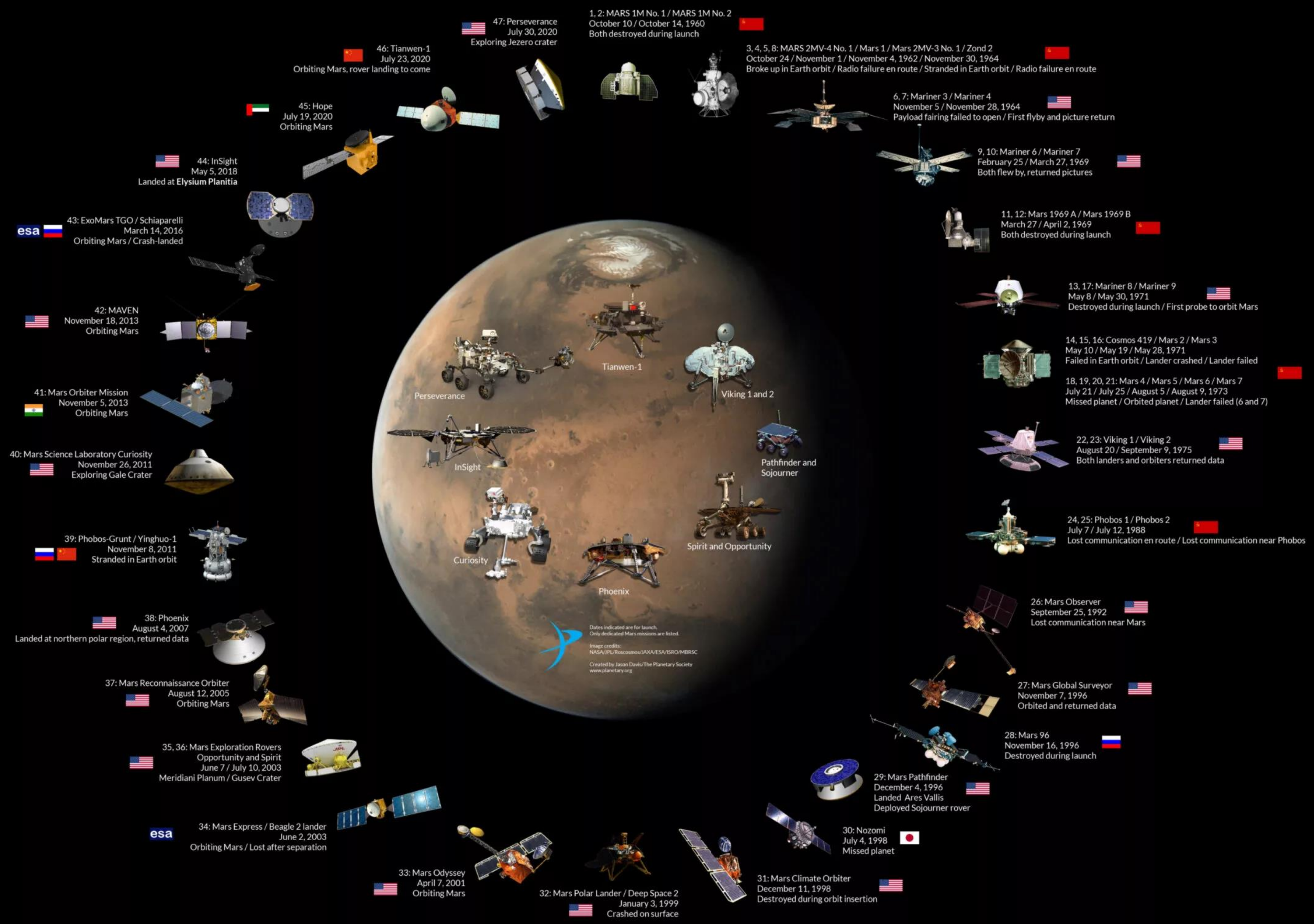
Schematic cross-section of the Moon showing the crust, mantle, and core layers inferred through analyses of the Apollo seismic data. The Apollo station locations and seismically active regions are indicated; arrowheads show a sample of meteoroid impact locations.

Figure reproduced from Nakamura, Latham, and Dorman (1982)

Topography Map of Mars



Mars Exploration Family Portrait

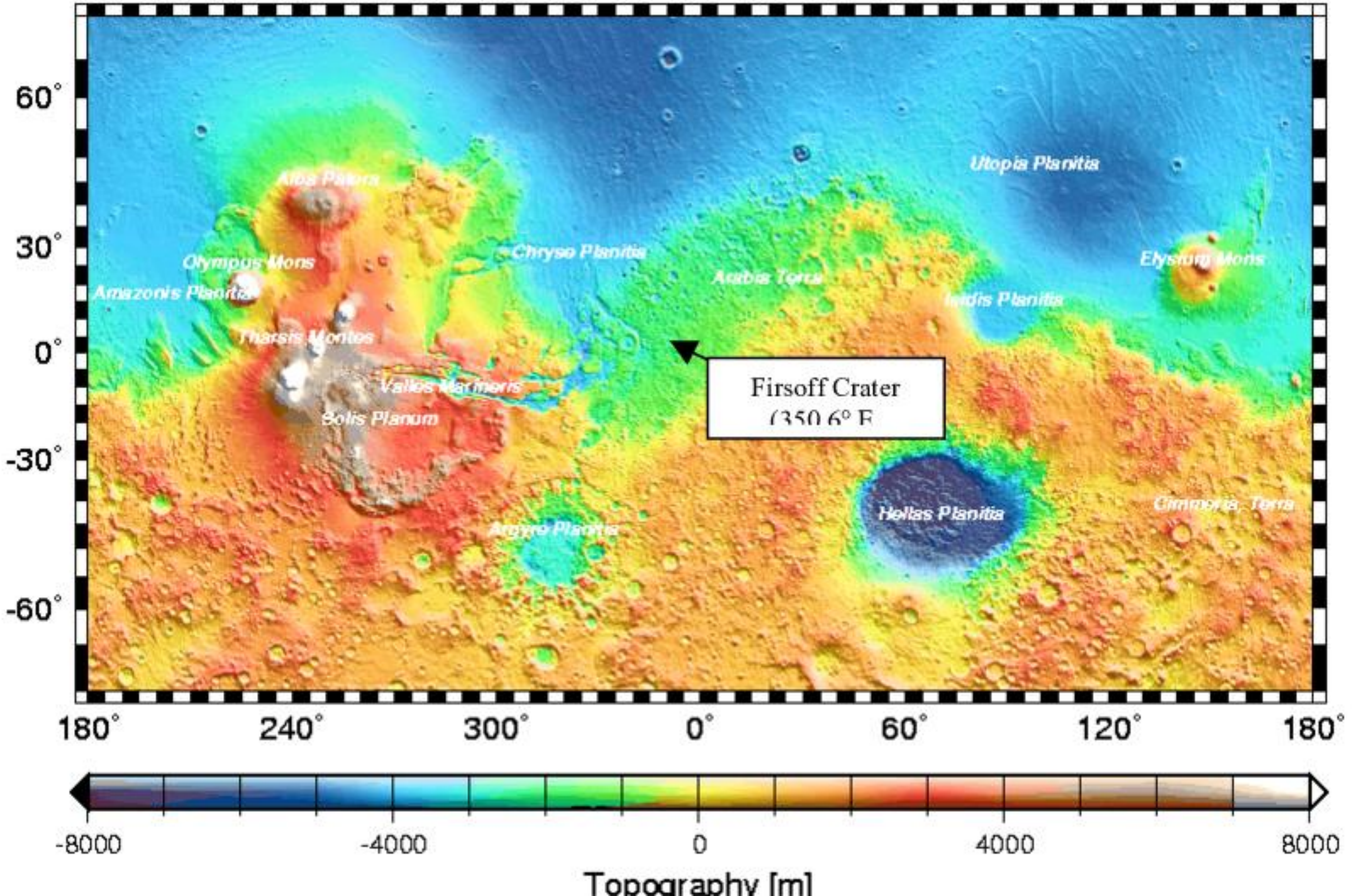


Various Mars Missions

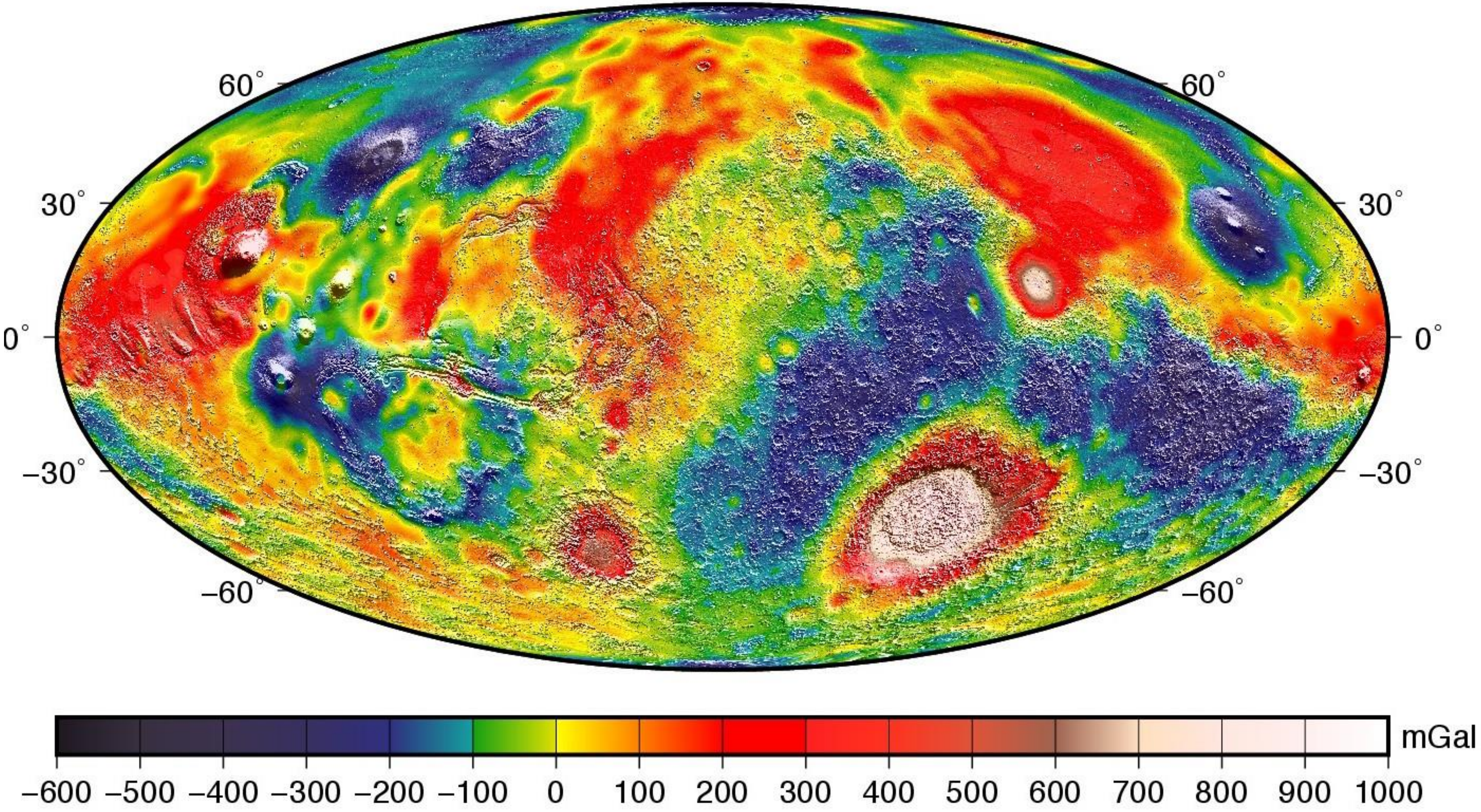
- ✓ **Mars 2020 Perseverance Rover**
- ✓ **InSight (Discovery Mission) 2018**
- ✓ **Mars Science Laboratory: Curiosity Rover, 2011**
- ✓ **Mars Phoenix, 2008**
- ✓ **Mars Exploration Rovers,** Rover Pair: "Spirit" and "Opportunity" **2008**
- ✓ **Mars Pathfinder, 1996**
- ✓ **Viking 1 & 2, 1975**

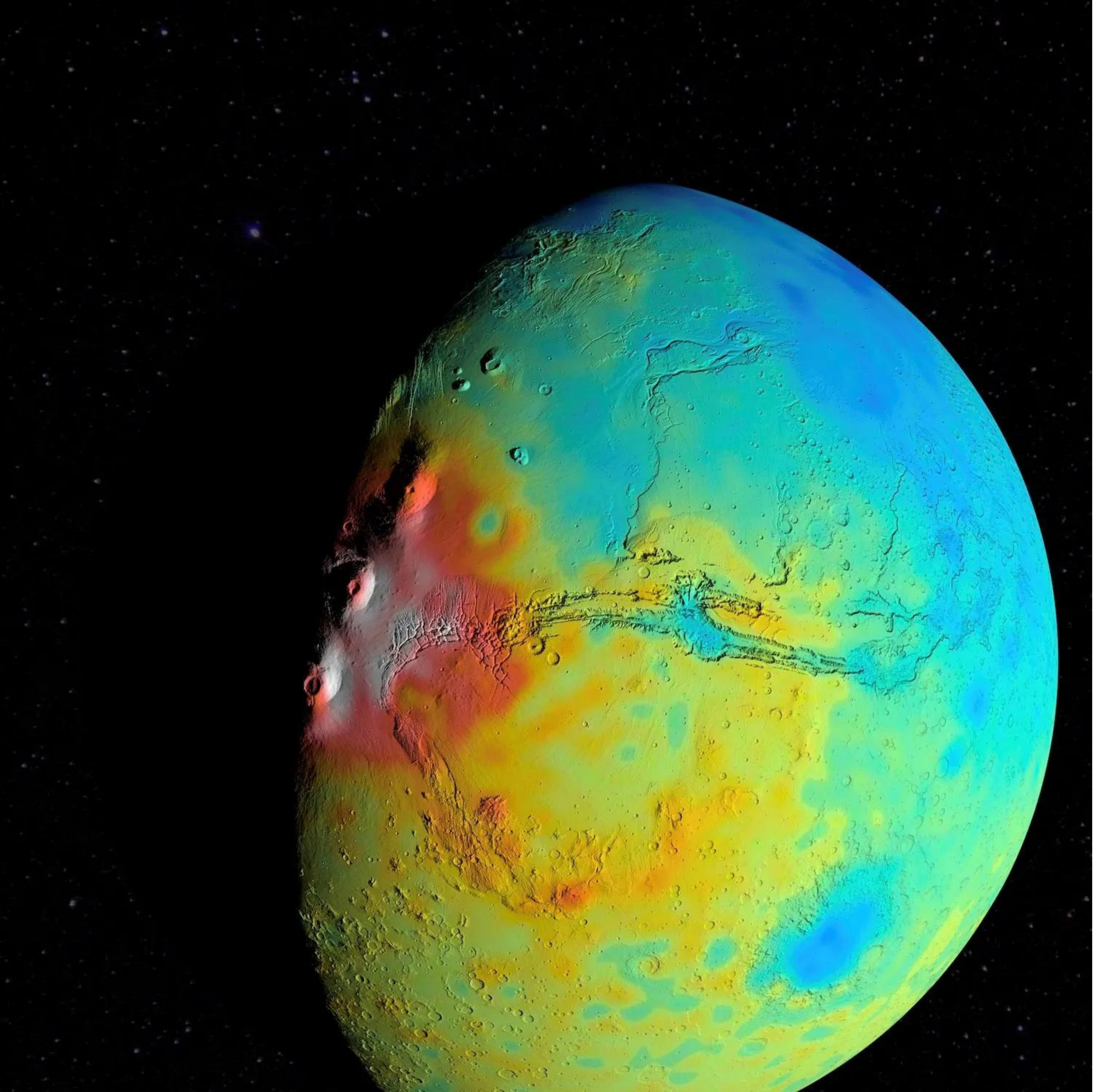
- ✓ **Mars Orbiter Mission, Mangalayaan 1, 2013**
- ✓ **Mangalayaan 2, 2023**

Topography Map of Mars



Gravity Anomaly Map of Mars





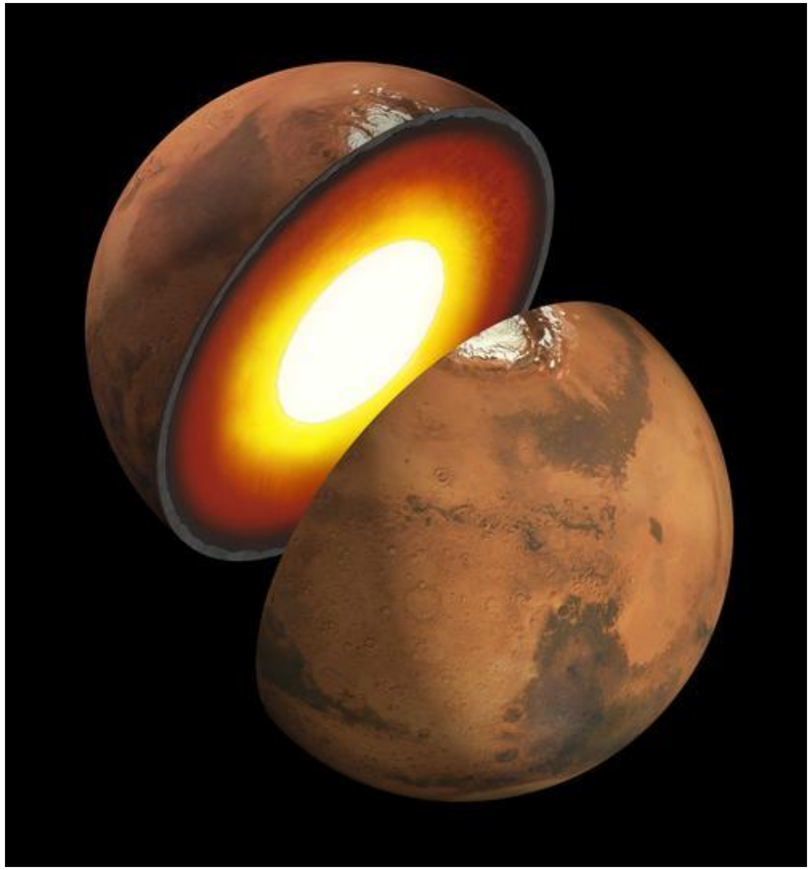
A new map of the thickness of Mars' crust shows less variation between thicker regions (red) and thinner regions (blue), compared to earlier mapping. This view is centered on Valles Marineris, with the Tharsis Montes near the terminator to its west. The map is based on modeling of the Red Planet's gravity field by scientists at NASA's Goddard Space Flight Center in Greenbelt, Maryland. The team found that globally Mars' crust is less dense, on average, than previously thought, which implies smaller variations in crustal thickness.

NASA/Goddard/UMBC/MIT/E. Mazarico

A rocky outcrop captured on Mar’s surface by Perseverance Rover

NASA'S MARS InSight Mission(2018)

- The InSight mission has two major goals, each with several science investigations, designed to help uncover the process that shaped all of the rocky planets in the inner solar system



To understand how rocky planets formed and evolved, InSight will study the interior structure and processes of Mars by determining:

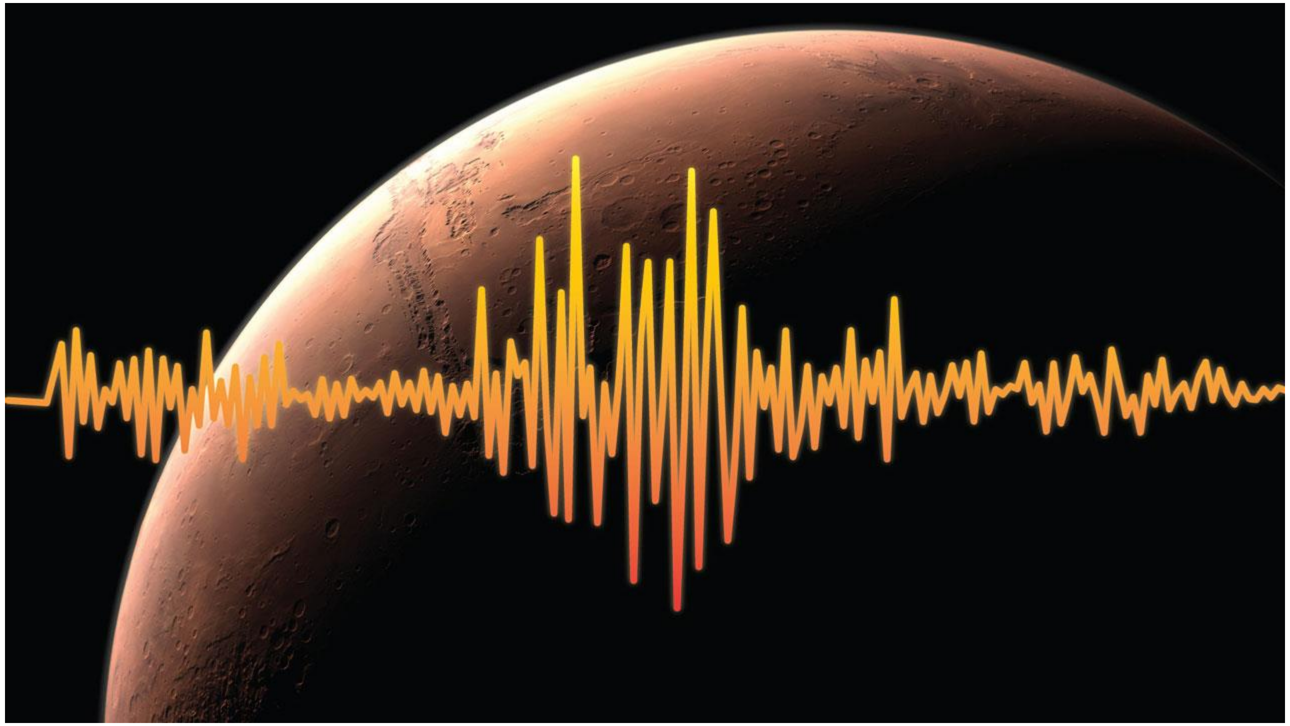
•The size of the core, what it is made of, and whether it is liquid or solid.

•The thickness and structure of the crust.

•The structure of the mantle and what it is made of.

•How warm the interior is and how much heat is still flowing through.

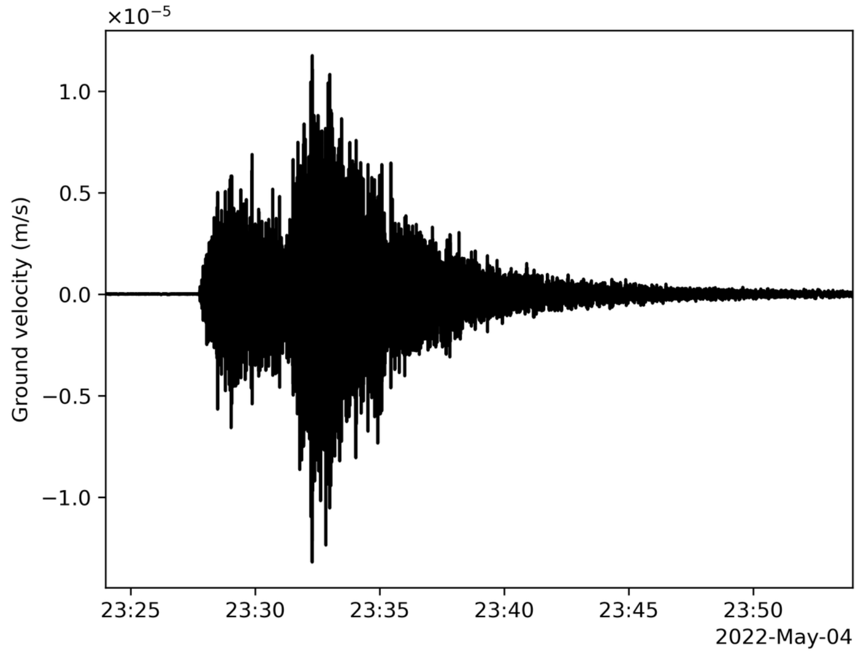
Insight will figure out just how tectonically active Mars is today, and how often meteorites impact it. For this, it will measure:

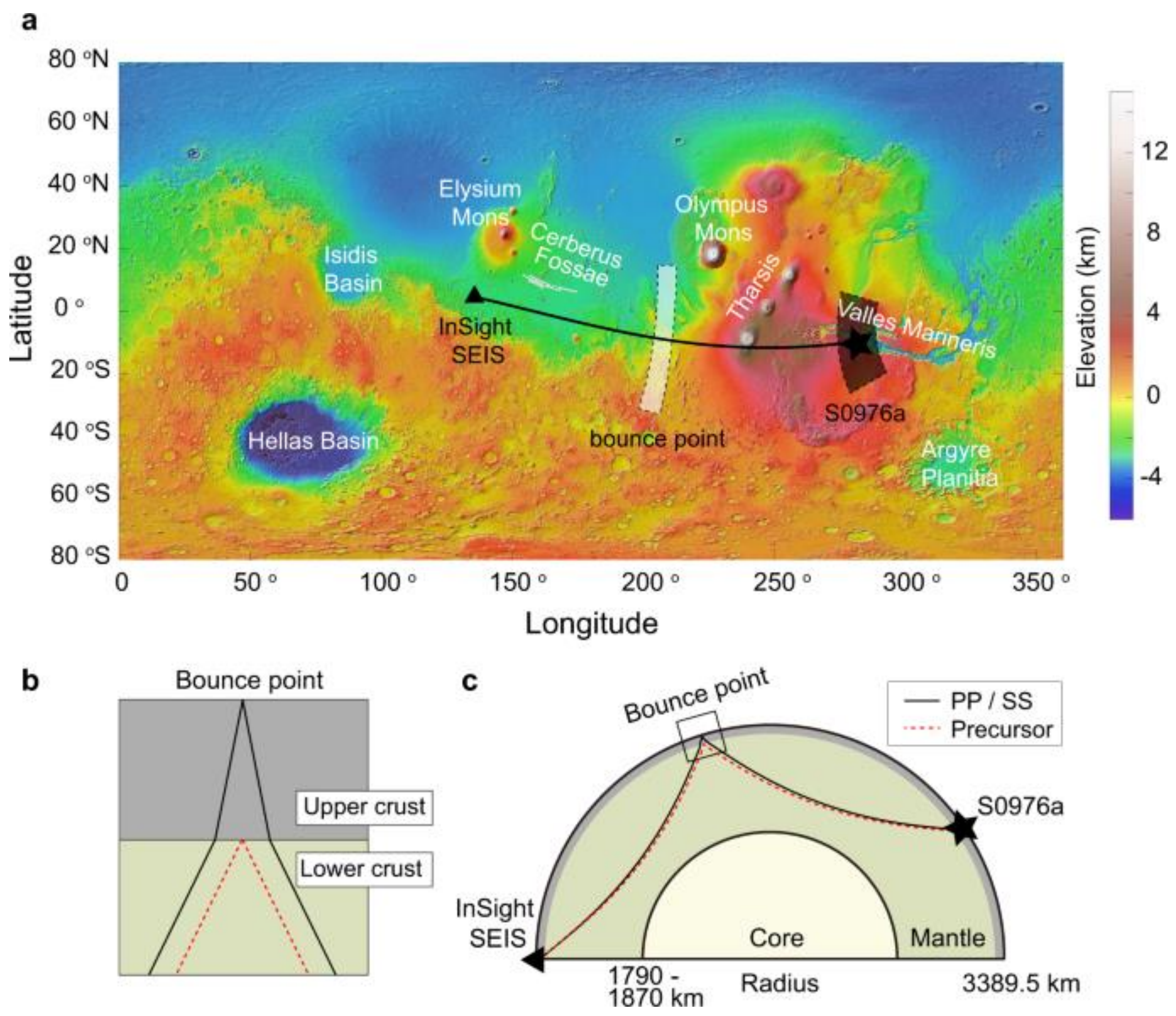


- How powerful and frequent internal seismic activity is on Mars, and where it is located within the structure of the planet.

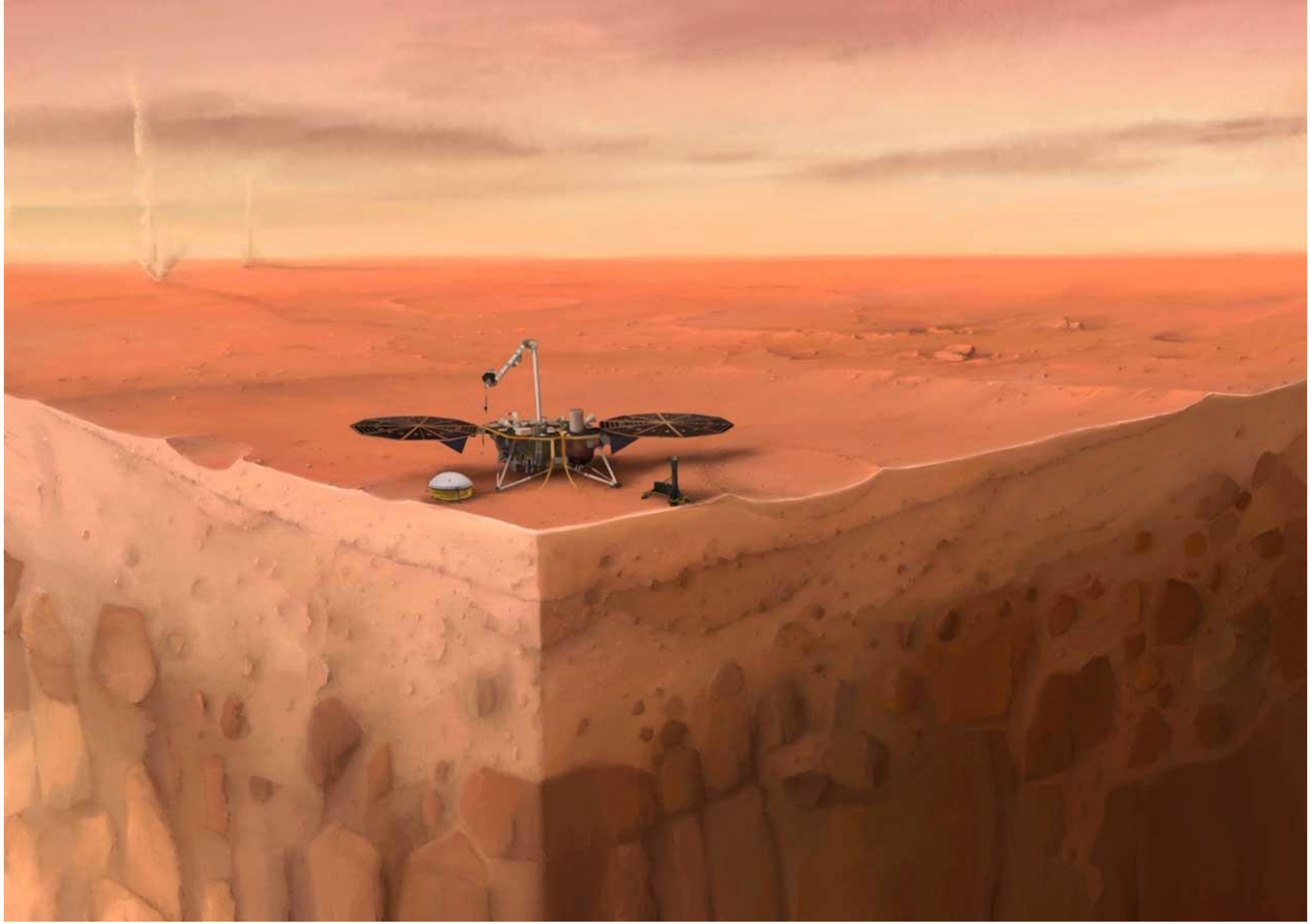
- How often meteorites impact the surface of Mars.

Seismicity on Mars

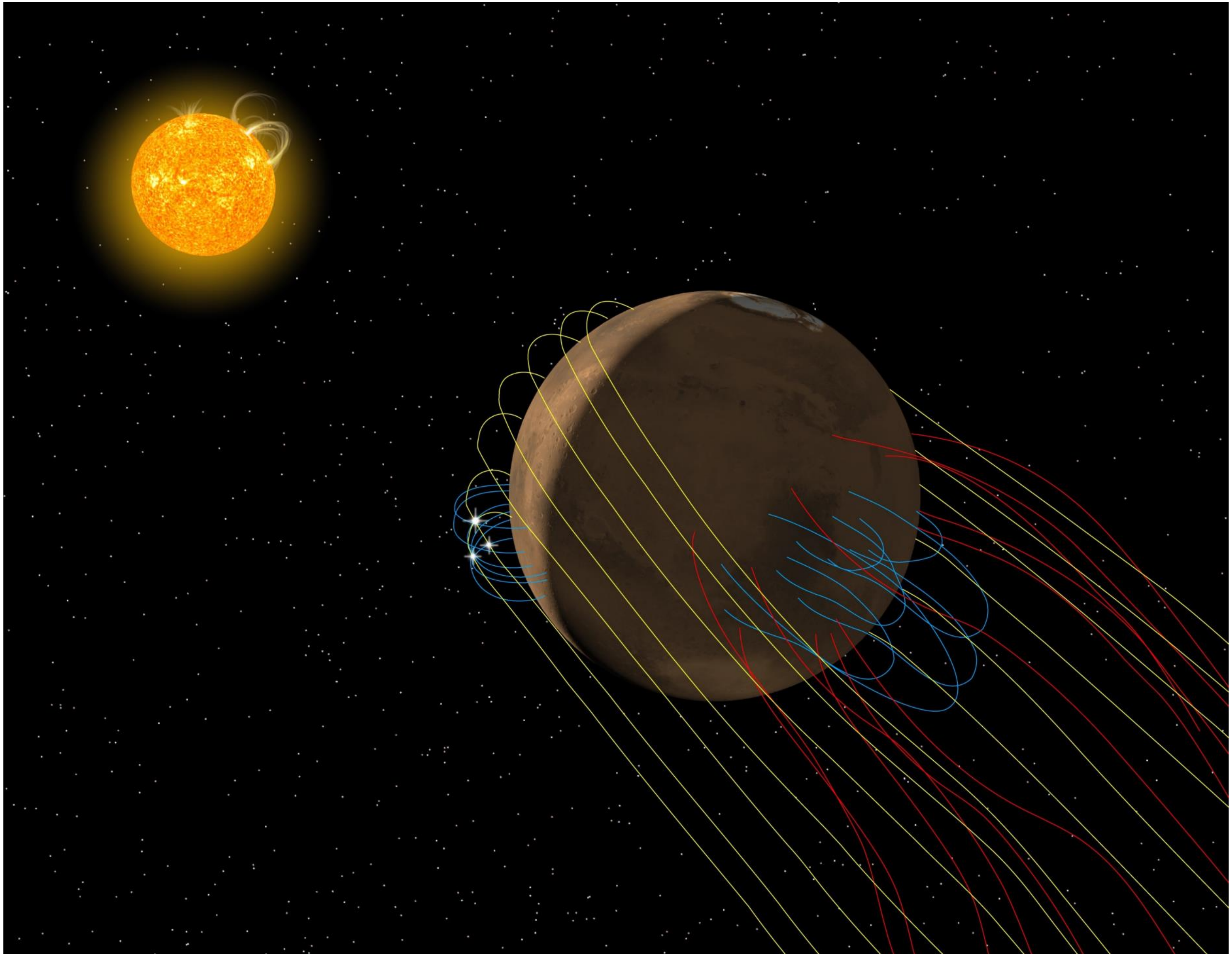




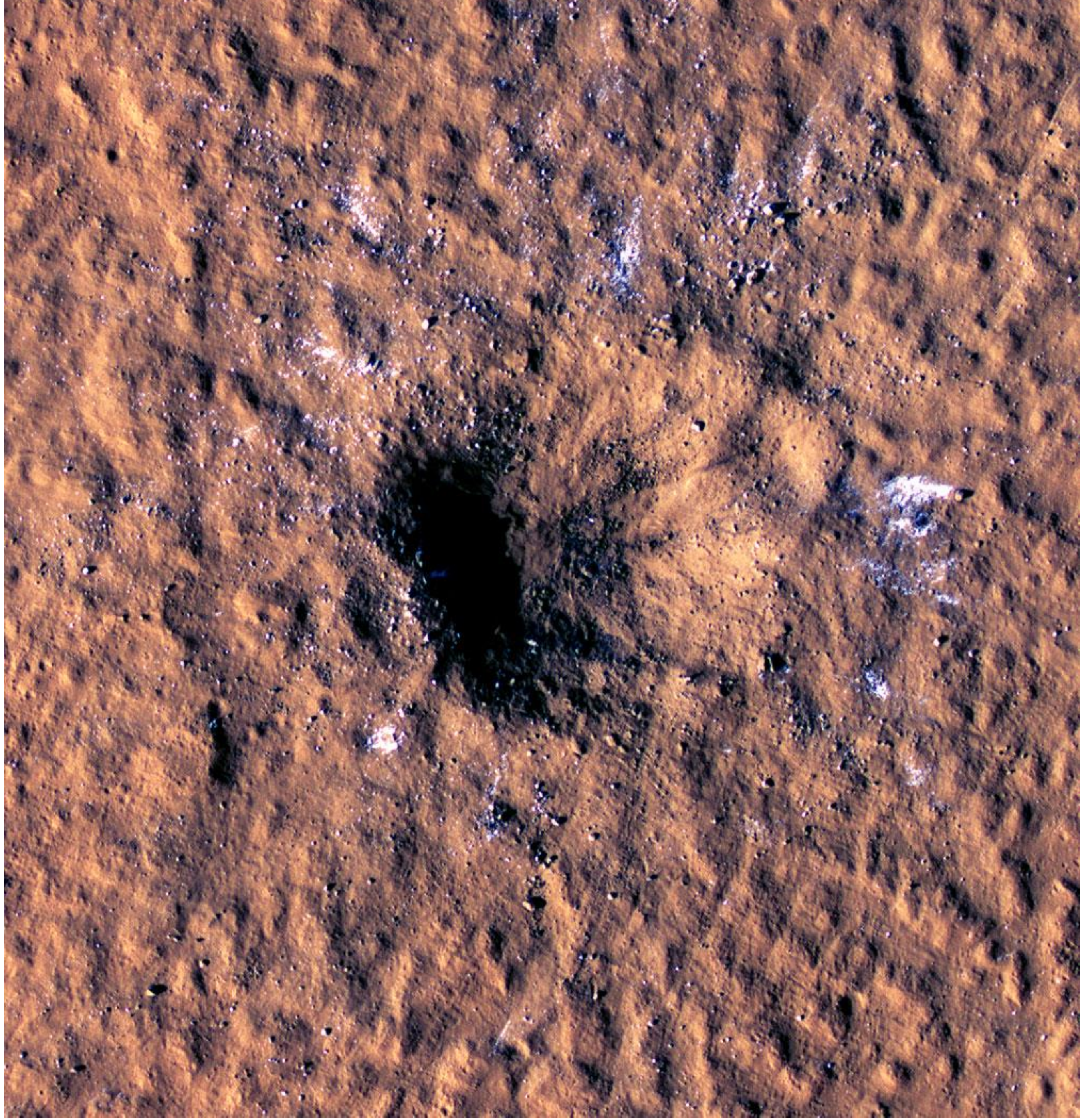
This seismogram shows the largest quake ever detected on another planet. Estimated at magnitude 5, this quake was discovered by NASA's InSight lander on May 4, 2022, the 1,222nd Martian day, or sol, of the mission.



A Cutaway of InSight on Mars: In this artist's concept of NASA's InSight lander on Mars, layers of the planet's subsurface can be seen below and dust devils can be seen in the background. Credits: IPGP/Nicolas Sarter



Artist's Conception of the Magnetic Field Environment at Mars: Artist's conception of the complex magnetic field environment at Mars.
Credits: Anil Rao/Univ. of Colorado/MAVEN/NASA GSFC



HiRISE Views a Mars Impact Crater Surrounded by Water Ice: Boulder-size blocks of water ice can be seen around the rim of this giant meteoroid impact crater on Mars, as viewed by the High-Resolution Imaging Science Experiment (HiRISE camera) aboard NASA's Mars Reconnaissance Orbiter. Credits: NASA/JPL-Caltech/University of Arizona